

Soc Internship day 11

Assignment:

- 1 Explain MITRE ATT&CK; in your own words
- 2 List 3 tactics and examples
- 3 Map a brute force attack to MITRE stages

(1)

MITRE ATT&CK helps Cybersecurity teams understand what the attacker is trying to do, how they are doing it, and what security team should detect or investigate. It is very useful in SOC, threat hunting, incident response, and detection engineering.

(2)

Tactic	Meaning	Example
Initial Access	How attacker enters the system	Phishing email with a malicious attachment
Credential Access	How attacker steals usernames/passwords	Using keylogger or password dumping tool
Exfiltration	How attacker steals data from the system	Uploading company files to attacker's cloud server

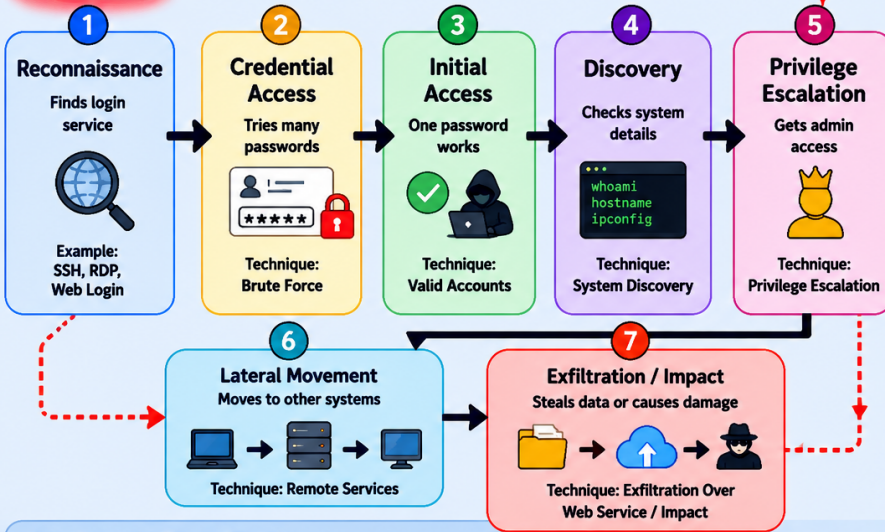
(3)

Attack Step	MITRE ATT&CK Tactic	MITRE Technique	Example
Attacker finds a login page or SSH/RDP service	Reconnaissance / Discovery	Active Scanning / Network Service Discovery	Finds SSH open on port 22
Attacker tries many passwords	Credential Access	Brute Force	Many failed login attempts for same user
Attacker successfully logs in	Initial Access	Valid Accounts	Login accepted using guessed password
Attacker checks system details	Discovery	System Owner/User Discovery	Runs <code>whoami</code> , <code>hostname</code> , <code>ipconfig</code>
Attacker tries to get admin access	Privilege Escalation	Valid Accounts / Exploitation	Uses weak admin password
Attacker moves to another system	Lateral Movement	Remote Services	Uses same password on another server
Attacker steals files	Exfiltration	Exfiltration Over Web Service	Uploads data outside the network

MITRE ATT&CK Mapping – Brute Force Attack



Attacker



MITRE ATT&CK Mapping	
Tactic	Technique
Reconnaissance	Active Scanning (Network Service Discovery)
Credential Access	Brute Force
Initial Access	Valid Accounts
Discovery	System Discovery
Privilege Escalation	Privilege Escalation (Valid Accounts)
Lateral Movement	Remote Services (SSH/RDP)
Exfiltration / Impact	Exfiltration Over Web Service / Impact

Log Example (Brute Force)

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10:17:45 Failed login for admin from 203.0.113.45
10:17:50 Failed login for admin from 203.0.113.45
10:17:55 Failed login for admin from 203.0.113.45
10:18:10 Accepted login for admin from 203.0.113.45
    
```

Many Failed Logins

Successful Login

Suspicious Commands

Access to Other Systems

Data Theft / Impact